Feature: UCRI-P2P — End-to-End IPL ↔ Adapter ↔ UCRM

Goal

Deliver **bidirectional UCRI 2.0 Peer-to-Peer** mission/event exchange between **IPL** and a **remote UCRM** via the **Adapter Integration**:

- **Forward flow:** IPL → Adapter → UCRM
- **Reverse flow:** UCRM → Adapter → IPL

This epic is intended to be delivered by a **cross-domain feature team** spanning **IPL-CORE**, **ADAPTER**, **INFRA**, and **TEST**.

Reference flow diagram

- UCRI 2.0 P2P FlowDiagram: UCRI 2.0- P2P (Peer-to-Peer) FlowDiagram

Feature Scope (what "done" means)

Forward flow (IPL → Adapter → UCRM)

1. IPL **constructs and sends** a TransferMission event to Adapter
2. ~~Adapter **validates** and returns **immediate validation response** to IPL ("accepted for processing" vs validation error)~~
3. Adapter **authenticates** to UCRM via P2P `/token`
4. Adapter **forwards** message to UCRM via `/messaging/send`
5. Adapter **interprets UCRM response** (accepted / rejected business / rejected technical / unknown)
6. Adapter **propagates final outcome** back to IPL, correlated to the original event

Reverse flow (UCRM → Adapter → IPL)

1. UCRM sends message via **same endpoint** `/messaging/send` to Adapter
2. Adapter **determines direction**, validates, and forwards to IPL
3. IPL returns accept/reject;
4. Adapter propagates outcome back to UCRM

Cross-cutting

1. End-to-end **logging/monitoring/traceability** for both directions using correlation IDs

Key epics in this scope:

- "UCRI 2: peer-to-peer": IPL-26313: UCRI 2: peer-to-peer OPEN

- “Event-based Adapter” (technical groundwork): IPL-26311: Event-based Adapter OPEN

Feature Team (cross domain)

Domains: IPL-CORE + ADAPTER + INFRA + TEST/QA

- **Epic Owner / PO** @JEHLE Mathias (single backlog owner for end-to-end outcome)
- **Tech Lead** @MADAAN Gautam (contracts, correlation, error handling)
- **Domain reps:** IPL-CORE, ADAPTER, INFRA, TEST/QA

Ownership boundaries

- **IPL-CORE** owns IPL payload construction, correlation storage, and ACK handling.
- **ADAPTER** owns validation, authentication, forwarding, response mapping, direction routing.
- **INFRA** owns runtime non-functionals: monitoring, alerting, secrets, deployment pipelines.

High-level RACI (cross-domain)

Legend: R = Responsible (does the work), **A** = Accountable (final decision/ownership), **C** = Consulted, **I** = Informed

Activity	PO	Tech Lead	IPL-CORE	ADAPTER	INFRA	TEST/QA	BA
1. Requirements & scope / backlog prioritization	**A**	C	C	C	C	C	**R/C**
1. Interface contract & message model decisions (incl. correlation IDs)	C	**A**	R/C	R/C	C	C	C
1. Payload mapping (IPL ↔ UCRI)	C	C	C	C	I	I	**R/A**
1. IPL implementation (send/receive/ACK + configuration)	I	C	**R/A**	C	C	C	C
1. Adapter implementation (validate, `/token`, `/messaging/send`,	I	C	C	**R/A**	C	C	C



routing, outcome mapping)							
1. Environment connectivity & certificates (OVH, Customer-ENV, routing/firewall/certs)	I	C	I	C	**R/A**	C	I
1. Observability (logging/metrics/alerts/ trace)	I	**A**	C	**R**	**R**	C	I
1. Test strategy & validation scenarios (OVH / pre-prod / customer env)	C	C	C	C	C	**R/A**	C
1. Release coordination & go-live readiness / runbook	**A**	C	R	R	R	R	I

Conflict resolution / decision rule: scope and priority conflicts are resolved by the **Epic Owner / PO**; technical alignment and architectural choices (contracts, correlation, error handling) are resolved by the **Tech Lead** after consulting the impacted domain teams.

Feature Milestones

Milestone (new order)	Milestone name	Suggested prefix	Notes	Completion date
1	Forward flow E2E against reference UCRM	`UCRI-P2P-M1-FWD`	"Forward" is the core baseline (IPL → Adapter → UCRM + outcome back).	TODO



2	Reverse flow + resilience + ops-ready baseline	`UCRI-P2P-M2-REVOPS`	Reverse + failure modes + ops/readiness items.	TODO
3	UCRM official Test Client available (quality gate)	`UCRI-P2P-M3-TC`	"UCRM Test Client" / reference impl readiness & connectivity gate.	TODO

Acceptance Criterias

- E2E **forward flow** operational; outcomes propagated to IPL
- E2E **reverse flow** operational; ACK loop complete
- **Configuration** available (tenant/dispatch-center enablement, partner routing)
- **Validation scenarios** executed (OVH / pre-prod / customer env)
- Monitoring endpoints + traceability (correlation ID) available

Included Stories (grouped by Domain / Team)

Business Analysts

Mapping

- ☑ IPL-26534: (ADAPTER) Mapping: IPL.TransferPatient ↔ UCRI.Patient OPEN
- ☑ IPL-26535: (ADAPTER) Mapping: IPL.TransferEinsatz ↔ UCRI.Incident OPEN

IPL-Core

Configuration:

- ☑ IPL-26528: (IPL-CORE) Tenant-Configuration (Mandatenkonfiguration UCRI) OPEN

Mock / Support Stories:

- ☑ IPL-26526: (IPL) IPL Sample Payloads (TransferEinsatz) OPEN

Functional specs:



- (IPL)User Story 1 – IPL Sends Event to Adapter
- (IPL)User Story 2 – IPL Handles Validation Response from Adapter
- (IPL)User Story 3 – IPL Processes “Accept” ACK from Adapter (UCRM Accepted)
- (IPL)User Story 4 – IPL Processes “Reject” ACK from Adapter (UCRM Rejected)
- (IPL)User Story 5 – IPL Processes Mission from UCRM via Adapter (Reverse Flow)
- (IPL)User Story 6 – IPL Sends ACK (Accept/Reject) for Mission to Adapter

Test

- IPL-26527: (IPL/ADAPTER) Validation-Scenarios for OVH / PRE-PROD / CUSTOMER-ENV OPEN
- IPL-26507: Integration Tests / End-To-End Tests / with IGNIS Plus Core Team on OVH OPEN

ADAPTER

UCRI P2P Adapter stories (Jira + Specs)

- (ADAPTER)User Story 1 – Receive Business (Tenant Payload) from IPL c:
- (ADAPTER)User Story 2 – Validate Incoming Request at Adapter
- (ADAPTER)User Story 3 – Authenticate Adapter to UCRM via P2P /token
- (ADAPTER)User Story 4 – Forward Message from Adapter to UCRM via /messaging/send
- (ADAPTER)User Story 5 – Handle UCRM Acceptance/Rejection Response at Adapter
- (ADAPTER)User Story 6 – Propagate UCRM Outcome Back to IPL
- (ADAPTER)User Story 7 – Support Reverse Flow UCRM → IPL via /messaging/send
- (ADAPTER)User Story 8 – Logging, Monitoring, and Traceability for P2P Flows

Mock / Resilience

- IPL-26525: (ADAPTER) UCRI-Mock IN PROGRESS
- IPL-26538: (ADAPTER) Resilinace Layer OPEN

INFRA

- IPL-26529: (INFRA) Testumgebung OVH mit UCRI-Kommunikationspartner (firewall & routing) OPEN
- IPL-26530: (INFRA) Certificate-Check Adapter <>UCRI-Partner OPEN
- **m**etrics/alerts/log correlation and operational monitoring
- secrets / secure config required

